DEBARUN GHOSH

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PROFESSIONAL SUMMARY

SRE and Generative AI engineer with hands-on experience in system design, platform & reliability engineering, and ML/data engineering for AIOps capabilities – agentic automation, LLM-driven operational workflows, and observability-integrated systems – in a production, globally distributed enterprise environment. Proven track record designing RAG pipelines, embeddings-based retrieval, agentic MCP workflows, and evaluation frameworks that reduce manual operational effort while maintaining reliability. Comfortable working across SRE, platform, and engineering teams to translate operational challenges into ML-based and automation-based solutions.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (Advanced), FastAPI, FastMCP, React.js, SQL, TensorFlow, Transformers
- **AIOps & Automation:** Agentic MCP Workflows, Splunk (Observability + AI Automation), Incident Automation, ChatOps-style Agent Design, Runbook Automation, Jenkins, Playwright
- **AI/ML & LLM Systems:** LLM Orchestration (Claude, Gemini), RAG Pipelines, Embeddings, LlamaIndex, Intent-driven Operational Chatbots, Prompt Engineering, Evaluation Datasets
- **Observability:** Splunk, Prometheus, Grafana, OTEL, Log/Metric/Trace-driven Automation Design
- **Streaming & Event-Driven:** Kafka (working knowledge), Pub/Sub (working knowledge)
- **Cloud & Platform:** GCP, Kubernetes, Microservices, Docker, Terraform, CI/CD, Distributed Data Processing, AlloyDB pgvector (Vector Indexing)

PROFESSIONAL EXPERIENCE

ANZ Operations and Technology Pvt. Ltd. — **Bengaluru, India**
Software Engineer – SRE & Generative AI — *May 2024 – Present*

- **AIOps – Intelligent Observability Automation (Splunk AI):** Architected an end-to-end agentic MCP workflow automating Splunk observability deployments across a distributed enterprise environment. Designed a dual-repository pattern decoupling the LLM orchestrator from execution tooling, reducing manual deployment effort from 2 days to 5 minutes per alert – a direct reduction in operational toil.
- **Agentic Governance Framework (Spec-Driven Development, Gemini CLI & Claude Code):** Built the Gemini CLI integration and its supporting MCP servers for an org-wide spec-driven development framework, enforcing spec-before-code discipline across AI-assisted engineering workflows. Designed multi-agent orchestration (planning, spec-writing, development, review) with automated gate checks, blocking unauthorized code writes and constraining agent execution to approved specifications – eliminating hallucinated or out-of-scope changes in AI-assisted workflows at enterprise scale.
- **Operational Knowledge Copilot (Enterprise Confluence Chatbot):** Designed and implemented a RAG-based operational copilot over 10,000+ enterprise documents, replacing manual/keyword search with semantic retrieval. Adopted by over a dozen internal teams for day-to-day operational queries with zero-hallucination, cited responses.
- **Automated Failure Mode Detection (Architecture Review Agent):** Built a 6-stage agentic evaluation pipeline (Gemini, AlloyDB pgvector) with 40+ curated hard-case scenarios and shadow-mode testing, proactively surfacing failure modes before they reached production review cycles.
- **Cloud Infrastructure & Security for AI/RAG Platform (GCP):** Provisioned and secured cloud-native infrastructure (139 commits, 30+ PRs, 1000+ LOC Terraform) across 8+ GCP services for an enterprise RAG/AI Hub platform. Led migration of services from Cloud Run to GKE for improved scalability and orchestration control. Implemented zero-trust OAuth/IAP authentication with automated

credential rotation via Secret Manager; built Pub/Sub event-streaming and Cloud Tasks async processing for agent orchestration; provisioned HA Redis and Cloud SQL layers; automated multi-environment CI/CD via GitHub Actions with least-privilege IAM. Decommissioned end-of-life services, reducing cloud footprint with zero downtime.

- **Cross-Functional Collaboration:** Partnered across SRE, platform, service management, and application engineering teams to translate operational and reliability challenges into ML-based and automation-first solutions, from initial design through production rollout.

RESEARCH & INDEPENDENT PROJECTS

- **Insight AI (Hardware-Aware LLM Deployment Profiler)**
  Problem Solved: Removed deployment guesswork by automatically scoring LLM deployment strategies against real system constraints.
  Tech Stack: Python, System Profiling APIs, Performance Tuning Algorithms.
  Execution: Built an engine analyzing CPU/RAM/GPU/VRAM headroom to rank deployment strategies by throughput and latency – directly applicable to capacity-aware AIOps automation.

- **AskDebarun (Conversational AI / Operational Chatbot Pattern)**
  Problem Solved: Built an interactive, intent-driven conversational agent pattern with session-based memory and API-level security.
  Tech Stack: LLaMA 3, ElevenLabs, FastAPI, NLP, RAG.
  Execution: Engineered a secure backend with CORS, input validation, and rate limiting – the same intent-model and chatbot design pattern applicable to ChatOps workflows.

- **Explainable AI for Enterprise Project Risk Prediction (Ongoing Research PoC)**
  Problem to be Solved: Bringing transparency to enterprise risk forecasting by grounding predictions in verifiable evidence rather than black-box scoring.
  Tech Stack: Python, Explainable AI (XAI) libraries, LLM Evaluation Frameworks.

EDUCATION

REVA University **July 2024**
B.Tech, Electronics & Communications Engineering

CERTIFICATIONS & PUBLICATIONS

- **Certifications:** Advanced Generative AI & Essentials of Generative AI (Purdue University Online), Python for Data Science (IBM)
- **Past Publications:** "Energy Efficiency in 5G Networks using GA & PSO" (IEEE NKCON '23); "Energy Consumption Saving in 5G Network Based on AI" (IEEE ICONAT '23)